



Limpets clamp themselves to rocks, protected by their thick conical shells

LIFE ON TIDAL SHORES

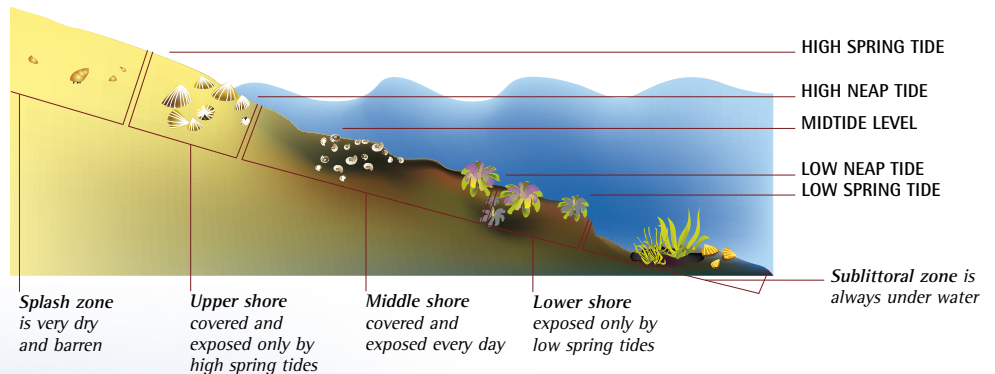
Water near the shore contains a lot of nutrients and drifting food, making the shoreline a rich habitat for marine animals. But it is also a very dangerous place. The breaking waves can crush any animal that is not concealed in crevices or burrows, and on tidal shores the falling tide exposes a region where attached animals such as mussels and barnacles risk drying up under the sun. The depth of this intertidal region changes every day, affecting what can live there, and this creates distinct shore zones inhabited by different types of marine life.

▲ DANGER AREA

As the waves crash on the shore, they pick up rocks and stones and toss them around. On rocky shores, animals can avoid the worst by creeping into crevices, which also prevents them from drying out as the tide falls. But most shore animals are very resilient, with extremely strong shells or tough rubbery skins.

SHORE ZONES ►

Different animals and seaweeds can survive different periods of exposure by the falling tide. The lowest parts of the shore are underwater nearly all the time, and are exposed only during low spring tides. The middle shore is covered and exposed every day, but the upper shore may be flooded for only a few hours twice a month, during high spring tides.



shoreline life



▲ LIVING WITH THE TIDES

Most tidal shore animals, such as these sea anemones, feed only when submerged. They survive low tide by closing up to retain water. Some do this more efficiently than others, so they can survive for longer and colonize higher zones of the shore.

◀ LIFE ZONES

Only a few types of animals can live above the lower shore, within tight limits imposed by the tide. But they multiply into dense colonies that form visible bands on rocky shores. On the far shore of this bay, massed mussels form a black band below zones of pale brown barnacles and yellow lichen.